Infosys Springboard Virtual Internship 6.0 Completion Report

Team Details

Batch Number: Batch 13

Start date:02/02/2026

Names:

1. Rinisha Shajitha
2. Om Bokare
3. Rinal Dave
4. Mihira Sri
5. Aditi
6. Alekhya
7. Rethiga

Internship Duration: 8 Weeks

1. Project Title

HotelRevAI AI Driven Revenue Analysis for Hotels

2. Project Objective

The objective of this project is to develop a comprehensive data analytics solution to analyze hotel booking data and generate actionable insights for improving business performance. The project focuses on evaluating key metrics such as occupancy rate, revenue generation, and customer behavior.

The system aims to assist hotel management in making data-driven decisions by identifying trends in bookings, cancellations, and customer preferences. It also helps in optimizing pricing strategies, improving occupancy rates, and enhancing customer satisfaction.

By leveraging Power BI, data modeling techniques, and interactive dashboards, the project transforms raw data into meaningful insights that support strategic planning and operational efficiency.

2. Project description in detail

The project focuses on developing a Hotel Revenue Optimization and Guest Analytics Dashboard using Power BI to analyze booking, customer, and room data. The approach began with data collection, cleaning, and transformation using Power Query to ensure accuracy and consistency.

A star schema data model was created with booking data as the fact table and Date, Customer, Room, and Hotel Branch as dimension tables. Calculated columns such as booking duration, stay type, and lead time were added to enhance analysis.

Key performance indicators like Occupancy Rate, Average Daily Rate (ADR), and Revenue Per Available Room (RevPAR) were developed using DAX to measure hotel performance.

The project was implemented through modules including occupancy and revenue analysis, guest segmentation, forecasting and cancellation trends, and a revenue strategy dashboard. Interactive dashboards with filters and visuals were created to present insights clearly.

Technologies used include Power BI, Power Query, DAX, and data modeling techniques.

This project has real-world impact by helping hotel management improve pricing strategies, optimize occupancy, reduce cancellations, and make data-driven decisions.

- ✓ **Module 1: Data Modeling and Ingestion**

The first phase of the project focused on data collection, cleaning, and modeling.

Datasets including booking details, customer information, and room data were imported into Power BI. Data preprocessing was performed using Power Query to handle missing values, remove duplicates, and standardize formats.

A **star schema model** was created to organize the data efficiently:

- Fact Table: Booking Data
- Dimension Tables: Date, Customer, Room, Hotel Branch

Relationships were established between tables to enable accurate filtering and aggregation.

Additional calculated columns were created:

- Booking Duration (difference between check-in and check-out dates)
- Room Category classification
- Stay Type (Weekend vs Weekday)
- Lead Time (time between booking and check-in)

This module served as the foundation for all further analysis.

- ✓ **Module 2: Occupancy & Revenue Metrics**

This module focused on evaluating hotel performance through key business metrics.

Using DAX, the following KPIs were created

- **Occupancy Rate (%)** to measure room utilization
- **Average Daily Rate (ADR)** to analyze revenue per occupied room
- **Revenue Per Available Room (RevPAR)** to evaluate overall revenue efficiency

Visualizations were developed to analyze:

- Daily, weekly, and seasonal performance trends
- Revenue distribution over time
- Comparison between direct bookings and OTA (Online Travel Agency) bookings

This module helped in identifying revenue patterns and performance gaps.

✓ **Module 3: Guest Analysis Module**

This module focused on analyzing customer behavior and segmentation.

Customers were categorized based on:

- Guest type (Business, Family, Solo travelers)
- Nationality and geographical distribution
- Booking source (Direct, OTA, etc.)

Advanced segmentation techniques were used to classify customers into:

- First-time guests
- Loyal customers
- High-spending customers

Visual dashboards were created to understand customer preferences, booking patterns, and stay duration. This module provides valuable insights for targeted marketing and customer retention strategies.

✓ **Module 4: Forecasting and Cancellation Trends**

This module focused on predictive analysis and identifying booking risks.

Trend analysis was performed using line charts to forecast future occupancy patterns. Key analyses included:

- Cancellation rate trends
- Lead time distribution

- No-show patterns
- Refund trends

These insights help in minimizing revenue loss and improving booking management strategies.

✓ **Module 5: Revenue Strategy Dashboard**

The final module integrates all insights into a strategic dashboard for decision-making.

Key features include:

- Identification of upselling opportunities (spa, dining, additional services)
- Seasonal pricing recommendations
- Room category performance analysis

An interactive dashboard was developed for hotel managers to monitor performance and make strategic decisions related to pricing, marketing, and operations.

✓ **Technologies Used**

- Power BI
- Power Query
- DAX (Data Analysis Expressions)
- Data Modeling (Star Schema)

✓ **Real-World Impact**

This project simulates real-world hotel analytics and provides insights that can:

- Improve revenue generation
- Optimize occupancy rates
- Reduce cancellations
- Enhance customer satisfaction
- Support strategic business decisions

3. Timeline Overview

Week	Activities Planned	Activities Completed
Week 1	Understanding requirements	Analyzed project scope and datasets
Week 2	Data cleaning	Cleaned and transformed data
Week 3	Data modeling	Built star schema and relationships
Week 4	KPI development	Created Occupancy %, ADR, RevPAR
Week 5	Dashboard creation	Built revenue and occupancy visuals
Week 6	Guest analysis	Developed segmentation dashboards
Week 7	Forecasting	Created trend and cancellation analysis
Week 8	Finalization	Completed dashboard and presentation

5a. Key Milestones

Milestone	Description	Date Achieved
Project Kickoff	Requirement analysis	Week 1
Prototype/First Draft	Initial dashboard	Week 4
Mid-Term Review	KPI completion	Week 6
Final Submission	Completed dashboard	Week 8
Presentation	Final demonstration	Week 8

5b. Project execution details

The project was executed in a systematic and structured manner, following a step-by-step analytical approach. Initially, raw datasets related to bookings, customers, and rooms were imported into Power BI and processed using Power Query. Data cleaning and transformation were performed to handle missing values, remove duplicates, and ensure consistency and accuracy across all fields.

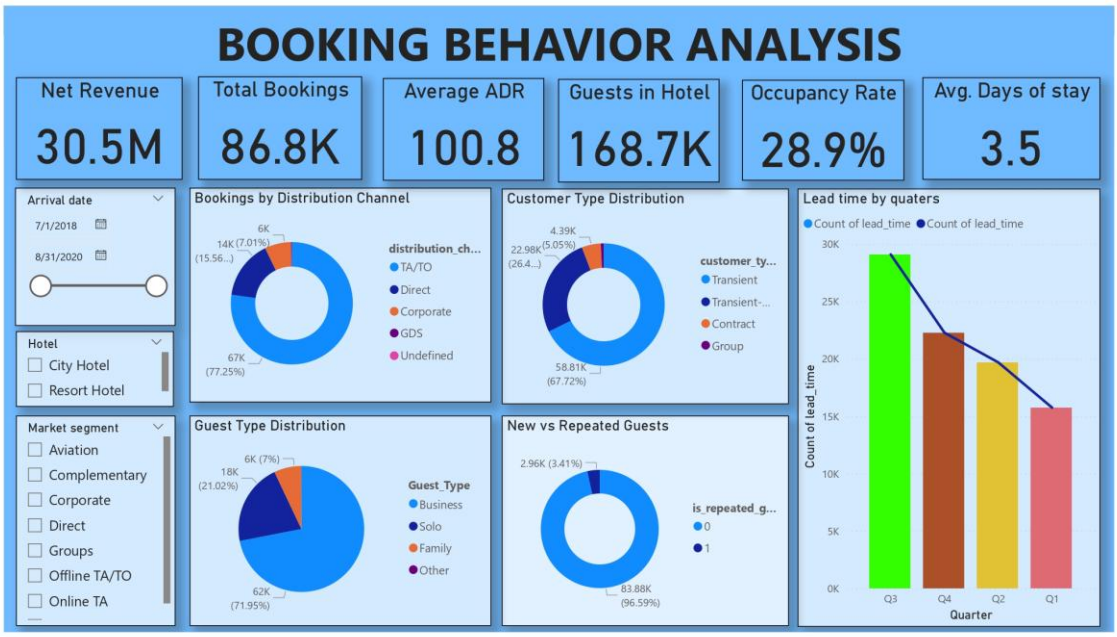
A star schema data model was then developed to organize the data efficiently, with booking data as the fact table and supporting dimension tables such as Date, Customer, Room, and Hotel Branch. Additional calculated columns like booking duration, stay type, and lead time were created to enhance analysis.

DAX formulas were used to calculate key performance metrics such as Occupancy Rate, Average Daily Rate (ADR), and Revenue Per Available Room (RevPAR), which formed the foundation for evaluating hotel performance.

Interactive dashboards were designed using Power BI visuals such as bar charts, line charts, and maps. Features like filters, slicers, and drill-down options were included to improve user interaction and enable dynamic data exploration.

Finally, trend analysis and forecasting techniques were applied to predict future occupancy patterns and analyze cancellation behavior. All insights were integrated into a comprehensive and user-friendly dashboard to support effective business decision-making.

6. Snapshots / Screenshots


BOOKING BEHAVIOR ANALYSIS
Net Revenue
30.5M
Total Bookings
86.8K
Average ADR
100.8
Guests in Hotel
168.7K
Occupancy Rate
28.9%
Avg. Days of stay
3.5
Arrival date
7/1/2018
8/31/2020
Hotel
City Hotel
Resort Hotel
Market segment
Aviation
Complementary
Corporate
Direct
Groups
Offline TA/TO
Online TA
Bookings by Distribution Channel
distribution_ch...
TA/TO
Direct
Corporate
GDS
Undefined
Customer Type Distribution
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Transient
Transient-...
Contract
Group
Lead time by quaters
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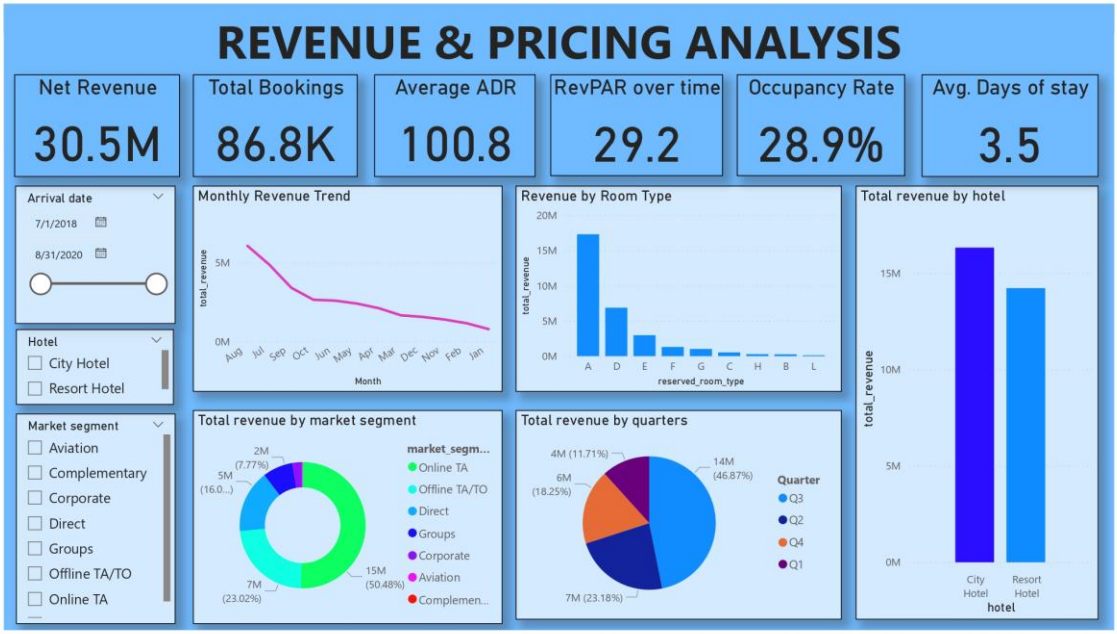
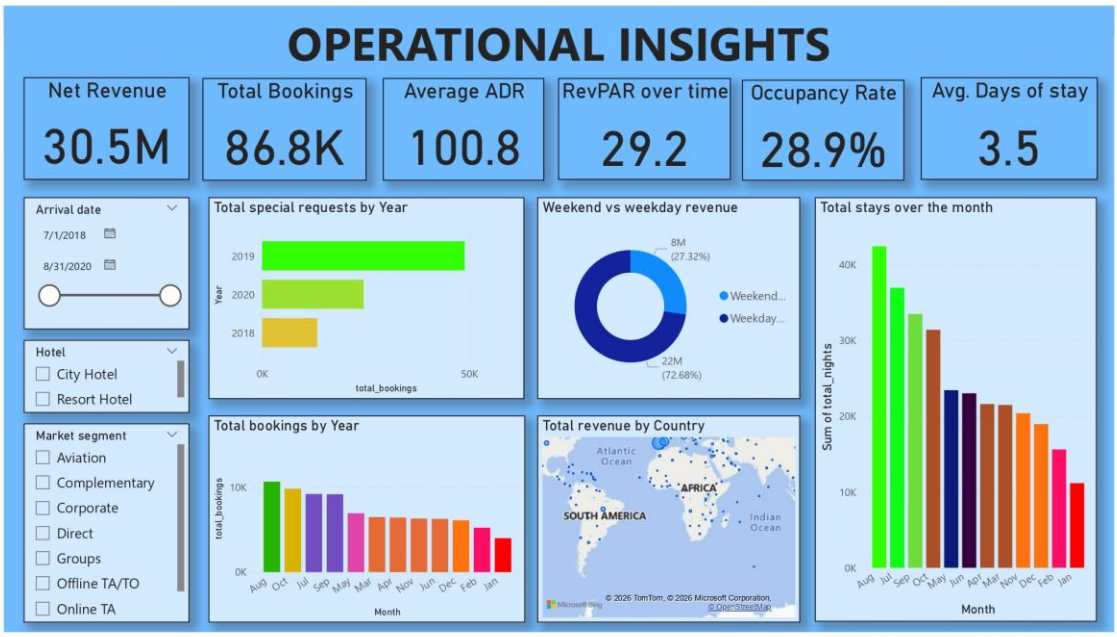
Quarter
Guest Type Distribution
Guest_Type
Business
Solo
Family
Other
New vs Repeated Guests
is_repeated_g...


REVENUE & PRICING ANALYSIS
Net Revenue
30.5M
Total Bookings
86.8K
Average ADR
100.8
RevPAR over time
29.2
Occupancy Rate
28.9%
Avg. Days of stay
3.5
Arrival date
7/1/2018
8/31/2020
Hotel
City Hotel
Resort Hotel
Market segment
Aviation
Complementary
Corporate
Direct
Groups
Offline TA/TO
Online TA
Monthly Revenue Trend
total_revenue
Month
Revenue by Room Type
reserved_room_type
Total revenue by hotel
hotel
Total revenue by market segment
market_segm...
Online TA
Offline TA/TO
Direct
Groups
Corporate
Aviation
Complemen...
Total revenue by quarters
Quarter
Q3
Q2
Q4
Q1



7. Challenges Faced

Several challenges were encountered during the project:

- Data inconsistency and missing values
- Difficulty in establishing proper table relationships
- Complexity in writing DAX formulas
- Sorting issues in time-based visuals
- Designing a professional and user-friendly dashboard

Solutions:

- Used Power Query for data cleaning
- Applied correct schema design
- Created helper columns for sorting
- Improved dashboard layout with consistent design principles

8. Learnings & Skills Acquired

This internship helped in gaining both technical and analytical skills.

Technical Skills:

- Power BI dashboard development
- Data modeling using star schema
- DAX for KPI calculations
- Data cleaning and transformation

Analytical Skills:

- Business data interpretation
- Customer segmentation
- Revenue and performance analysis

Soft Skills:

- Problem-solving
- Time management
- Communication and presentation skills

9. Testimonials from team

This internship provided a valuable opportunity for our team to work collaboratively on a real-world data analytics project, enabling us to apply our theoretical knowledge in a practical and industry-relevant environment. Throughout the project, we gained hands-on experience in handling real datasets and understanding how data-driven insights can support business decision-making. It significantly enhanced our understanding of data visualization, data modeling, and business intelligence tools, particularly Power BI.

As a team, we were involved in every stage of the project, including data collection, cleaning, transformation, and analysis. We developed the ability to work with complex datasets, identify patterns, and derive meaningful insights. By building interactive dashboards, we learned how to effectively present data in a clear and visually appealing manner, making it easier for stakeholders to interpret and utilize the information.

The project also helped us strengthen important soft skills such as teamwork, communication, and time management. Collaborating as a team allowed us to share ideas, solve problems collectively, and improve the overall quality of our work. Additionally, the experience enhanced our problem-solving abilities and boosted our confidence in handling data and presenting insights.

Overall, this internship has been a highly enriching experience for our team, equipping us with both technical and analytical skills, and preparing us for future roles in data analytics and business intelligence.

10. Conclusion

In conclusion, this project successfully demonstrated the practical application of data analytics in improving hotel business performance and decision-making. The developed dashboard provides clear and meaningful insights into key areas such as occupancy trends, revenue generation, and customer behavior, enabling better strategic planning.

Overall, this internship experience has significantly strengthened my technical, analytical, and problem-solving skills. It has also enhanced my understanding of business intelligence tools and prepared me for future roles in data analytics and business intelligence domains.

11. Acknowledgements

We would like to express our sincere gratitude to Infosys Springboard for providing us with this valuable internship opportunity. This experience has allowed us to gain practical exposure and enhance our knowledge in the field of data analytics and business intelligence.

We would also like to extend our heartfelt thanks to our mentors for their continuous guidance, support, and valuable feedback throughout the project. Their insights played a crucial role in helping us successfully complete our work.

Additionally, we thank all our team members for their collaboration, coordination, and collective efforts. Working together as a team helped us overcome challenges, share knowledge, and achieve our project goals effectively.